

Lecture 30/Module 3

Sinusoidal Steady State Analysis

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Formulas Revision – AC Power

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$

1. Instantaneous Power

$$p(t) = v(t)i(t)$$



$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

2. Average Power

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \end{aligned}$$

3. Apparent Power

$$S = V_{\text{rms}} I_{\text{rms}}$$

4. Power Factor

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

4. Complex Power

$$\begin{aligned} S &= V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i) \end{aligned}$$

Real Power

Reactive Power

Note: Expressions for Average Power and Real Power are same

Problem 1

The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ)$ V and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ)$ A. Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Solution 1

(a) For the rms values of the voltage and current, we write

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}} \angle -10^\circ, \quad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}} \angle +50^\circ$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}} \angle -10^\circ \right) \left(\frac{1.5}{\sqrt{2}} \angle -50^\circ \right) = 45 \angle -60^\circ \text{ VA}$$

The apparent power is

$$S = |\mathbf{S}| = 45 \text{ VA}$$

Solution 1

(b) We can express the complex power in rectangular form as

$$S = 45 \angle -60^\circ = 45[\cos(-60^\circ) + j \sin(-60^\circ)] = 22.5 - j38.97$$

Since $S = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$Q = -38.97 \text{ VAR}$$

(c) The power factor is

$$\text{pf} = \cos(-60^\circ) = 0.5 \text{ (leading)}$$

It is leading, because the reactive power is negative. The load impedance is

$$Z = \frac{V}{I} = \frac{60 \angle -10^\circ}{1.5 \angle +50^\circ} = 40 \angle -60^\circ \Omega$$

which is a capacitive impedance.

Problem 2

A load $\mathbf{Z}$ draws 12 kVA at a power factor of 0.856 lagging from a 120-V rms sinusoidal source. Calculate: (a) the average and reactive powers delivered to the load, (b) the peak current, and (c) the load impedance.

Solution 2

(a) Given that $\text{pf} = \cos\theta = 0.856$, we obtain the power angle as $\theta = \cos^{-1} 0.856 = 31.13^\circ$. If the apparent power is $S = 12,000$ VA, then the average or real power is

$$P = S \cos\theta = 12,000 \times 0.856 = 10.272 \text{ kW}$$

while the reactive power is

$$Q = S \sin\theta = 12,000 \times 0.517 = 6.204 \text{ kVA}$$

(b) Since the pf is lagging, the complex power is

$$S = P + jQ = 10.272 + j6.204 \text{ kVA}$$

$$I_m = \sqrt{2}I_{\text{rms}} = \sqrt{2}(100) = 141.4 \text{ A}$$

From $S = V_{\text{rms}} \mathbf{I}_{\text{rms}}^*$, we obtain

$$\mathbf{I}_{\text{rms}}^* = \frac{S}{V_{\text{rms}}} = \frac{10,272 + j6,204}{120 \angle 0^\circ} = 85.6 + j51.7 \text{ A} = 100 \angle 31.13^\circ \text{ A}$$



Solution 2

(c) The load impedance

$$\mathbf{Z} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{120 \angle 0^\circ}{100 \angle -31.13^\circ} = 1.2 \angle 31.13^\circ \Omega$$

which is an inductive impedance.

Problem 3

In the circuit of Fig. 11.26, $\mathbf{Z}_1 = 60 \angle -30^\circ \Omega$ and $\mathbf{Z}_2 = 40 \angle 45^\circ \Omega$. Calculate the total: (a) apparent power, (b) real power, (c) reactive power, and (d) pf, supplied by the source and seen by the source.

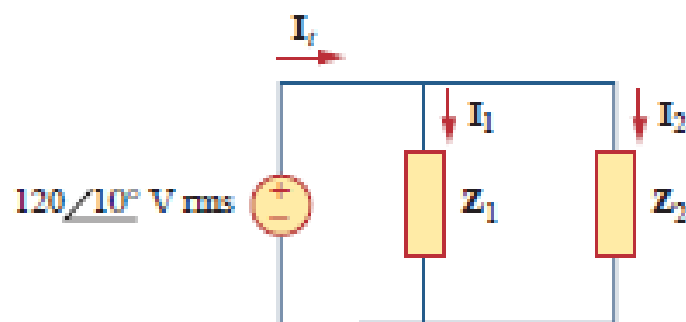


Figure 11.26
For Example 11.14.

Solution 3

The current through Z_1 is

$$I_1 = \frac{V}{Z_1} = \frac{120\angle 10^\circ}{60\angle -30^\circ} = 2\angle 40^\circ \text{ A rms}$$

while the current through Z_2 is

$$I_2 = \frac{V}{Z_2} = \frac{120\angle 10^\circ}{40\angle 45^\circ} = 3\angle -35^\circ \text{ A rms}$$

The complex powers absorbed by the impedances are

$$S_1 = \frac{V_{rms}^2}{Z_1^*} = \frac{(120)^2}{60\angle 30^\circ} = 240\angle -30^\circ = 207.85 - j120 \text{ VA}$$

$$S_2 = \frac{V_{rms}^2}{Z_2^*} = \frac{(120)^2}{40\angle -45^\circ} = 360\angle 45^\circ = 254.6 + j254.6 \text{ VA}$$

The total complex power is

$$S_t = S_1 + S_2 = 462.4 + j134.6 \text{ VA}$$

(a) The total apparent power is

$$|S_t| = \sqrt{462.4^2 + 134.6^2} = 481.6 \text{ VA}$$



(b) The total real power is

$$P_t = \text{Re}(S_t) = 462.4 \text{ W or } P_t = P_1 + P_2.$$

(c) The total reactive power is

$$Q_t = \text{Im}(S_t) = 134.6 \text{ VAR or } Q_t = Q_1 + Q_2.$$

(d) The pf = $P_t/|S_t| = 462.4/481.6 = 0.96$ (lagging).

We may cross check the result by finding the complex power S_s supplied by the source.

$$\begin{aligned} I_t &= I_1 + I_2 = (1.532 + j1.286) + (2.457 - j1.721) \\ &= 4 - j0.435 = 4.024\angle -6.21^\circ \text{ A rms} \end{aligned}$$

$$\begin{aligned} S_s &= VI_t^* = (120\angle 10^\circ)(4.024\angle 6.21^\circ) \\ &= 482.88\angle 16.21^\circ = 463 + j135 \text{ VA} \end{aligned}$$

which is the same as before.

Thank You